Railway Fare and Reservation Calculator

Complete worked solution

Sample fare policy: base fare per km is 1.00 (standard), 1.5 times for AC, and 0.75 times for sleeper. For each passenger, base fare is charged by distance slabs: first 100 km at full class rate, next 200 km at 80%, and distance above 300 km at 60%. Children under 5 travel free, ages 5-11 receive 50% fare, seniors 60+ receive 25% concession. Reservation fee is 40 per passenger. Cancellation penalty is 10% of fare before reservation fee, with minimum 50 and maximum equal to the amount paid.

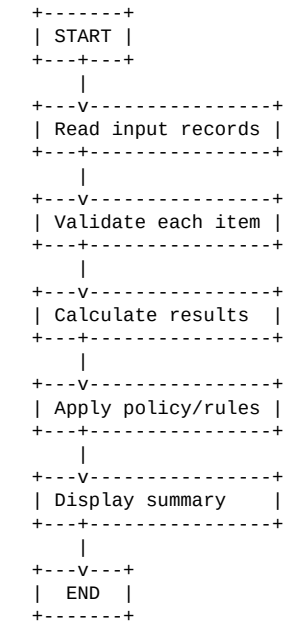
1. Computational Thinking and Algorithm

Inputs: Supplied records and policy values. **Processing:** Validate data, calculate charges or scores, apply ordered rules, and compare outcomes. **Outputs:** Itemized results, classifications, rankings, or recommendations.

Pseudocode

```
START
  Read trip distance, class, passengers, payment, cancellation flag
  Determine class multiplier
  Calculate distance slab fare
  FOR each passenger
    Validate age
    Apply age concession or free travel rule
    Add reservation fee if fare is chargeable
  Calculate total booking cost
  IF cancelled, compute bounded cancellation penalty and refund
  Display passenger fare breakdown and total
END
```

2. Flowchart



3. Python Program

The program uses basic variables, numeric and string data, tuples, lists, loops, conditionals, arithmetic, and formatted output. Policy values are constants near the top of the program.

```
passengers = [("Anu", 32), ("Kavi", 8), ("Mohan", 66)]
distance_km = 420
travel_class = "AC"  # valid: STANDARD, AC, SLEEPER
```

```
paid_amount = 2200
cancel_booking = True
class_multiplier = 1.0
if travel_class == "AC":
    class_multiplier = 1.5
elif travel_class == "SLEEPER":
    class_multiplier = 0.75
elif travel_class != "STANDARD":
    print("Invalid class")
    class_multiplier = -1

if distance_km < 0 or class_multiplier < 0:
    print("Invalid trip")
else:
    rate = class_multiplier
    if distance_km <= 100:
        distance_fare = distance_km * rate
    elif distance_km <= 300:
        distance_fare = 100 * rate + (distance_km - 100) * rate * 0.80
    else:
        distance_fare = (100 * rate + 200 * rate * 0.80
                         + (distance_km - 300) * rate * 0.60)
    booking_total = 0
    for name, age in passengers:
        if age < 0 or age > 120:
            print("Invalid age for", name)
            continue
        if age < 5:
            fare = 0
        elif age <= 11:
            fare = distance_fare * 0.50
        elif age >= 60:
            fare = distance_fare * 0.75
        else:
            fare = distance_fare
        reservation = 40 if fare > 0 else 0
        passenger_total = fare + reservation
        booking_total = booking_total + passenger_total
        print(name, "age", age, "fare", round(fare, 2),
              "reservation", reservation, "total", round(passenger_total, 2))
    if cancel_booking:
        penalty = max(50, booking_total * 0.10)
        penalty = min(penalty, paid_amount)
        refund = paid_amount - penalty
        print("Cancellation penalty:", round(penalty, 2),
              "Refund:", round(refund, 2))
    else:
        print("Booking total:", round(booking_total, 2),
              "Amount due:", round(max(0, booking_total - paid_amount), 2))
```

4. Sample run guidance

Run the program as written to see a sample result. Replace the sample lists and constants with the values provided in your assignment. Keep the rule order and units consistent; for example, percentages should be entered as percentages and converted to fractions only in the calculation.